

Question	Answer	Marks	Guidance
3(a)	(A wave which) transfers energy (in the direction of propagation)	B1	Allow 'it' to mean 'a wave' provided not clearly referring to something else Allow 'propagates / carries / moves' for transfers Ignore references to medium / matter Ignore references to transverse or longitudinal waves
3(b)(i)	$T = 0.33 \text{ s}$	A1	
3(b)(ii)	$\lambda = 3.9 - 2.4$ $= 1.5$	C1	The C1 marks may be credited independently of each other
	$v = f\lambda$ or $v = \lambda / T$ $= 3 \times 1.5$ $= 1.5 / 0.33$	C1	Allow $c = f\lambda$ ECF T from 3(b)(i)
	$= 4.5 \text{ m s}^{-1}$	A1	

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3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1	
	$= 0.44 \times 9.81 \times (0.15 \times 2)$	C1	Allow $g = 10$ as leads to same answer to 2SF Allow $0.44 \times 9.81 \times (-0.15 \times 2)$ $0.44 \times 9.81 \times 0.15$ implies 1st C1 mark but not 2nd C1 mark
	$= 1.3 \text{ J}$	A1	Allow -1.3 J
3(c)(ii)	The path difference (from A and B) is not a whole number of wavelengths / (The waves at P are) not in phase	B1	Allow '(waves are) out of phase' / 'phase difference not zero' Not '(waves are) 180° out of phase / antiphase' Allow 'not (complete) constructive interference (at P)' (BOD)
	(The amplitude) decreases	B1	Allow 'it' to mean amplitude (unless it clearly refers to something else) Allow 'less' / 'smaller' for decreases